19.5-Lab: Configure and Secure SSH

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```
student@workstation:~$ lab start ssh-review
🤖 Running: lab start ssh-review
SUCCESS Ensuring the required packages are installed on servera
SUCCESS Backing up the /etc/ssh/sshd_config file on serverb
SUCCESS Adding the production1 user on servera
SUCCESS Adding the production1 user on serverb
SUCCESS Adding the production2 user on serverb
SUCCESS Permitting SSH as root on serverb
SUCCESS Handling password authentication on serverb
student@workstation:~$ ssh student@servera
Web console: https://servera.lab.example.com:9090/ or https://172.25.250.10:9090/

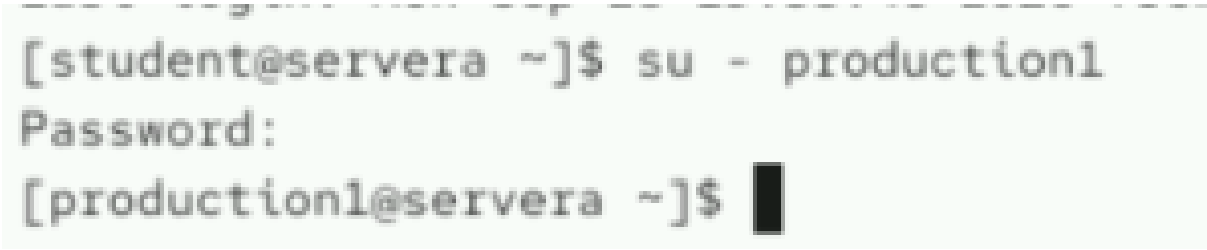
Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Mon Sep 28 19:55:46 2026 from 172.25.250.9
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```
[production1@servera ~]$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/production1/.ssh/id_ed25519):
Created directory '/home/production1/.ssh'.
Enter passphrase for "/home/production1/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/production1/.ssh/id_ed25519
Your public key has been saved in /home/production1/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:jlIgbiYfg1Mo7cFrtFczaZ500iMiVRtxHTn1L/AsoaI production1@servera
The key's randomart image is:
+--[ED25519 256]--+
|    ..+....+.    |
| + .   *  +  .   |
|o O o @ +  .o .  |
|.* * B B . . = . |
|+ @ . + S . . + .|
| B + . + .   . . |
|  . . E .        |
|     .           |
|                 |
+----[SHA256]-----+
[production1@servera ~]$
```

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```
[production1@servera ~]$ ssh-copy-id production1@serverb
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/production1/.ssh/id_ed25519.pub"
The authenticity of host 'serverb (172.25.250.11)' can't be established.
ED25519 key fingerprint is SHA256:iS9Q0/bXHOwXKX5zTP10nAIxSnKQoUQxMfKE0s0b2ZU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
production1@serverb's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'production1@serverb'"
and check to make sure that only the key(s) you wanted were added.

[production1@servera ~]$
```

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```
[production1@servera ~]$  ssh production1@serverb
Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

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management capabilities on your system.
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You can learn more about how to register your system
using rhc at https://red.ht/registration
[production1@serverb ~]$
```

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```
[production1@serverb ~]$ su -
Password:
[root@serverb ~]# vim /etc/ssh/sshd_config
```

```
#LoginGraceTime 2m
PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes
"/etc/ssh/sshd_config" 131L, 3726B
```

```
[root@serverb ~]# systemctl restart sshd.service
[root@serverb ~]# ssh production1@servera
The authenticity of host 'servera (172.25.250.10)' can't be established.
ED25519 key fingerprint is SHA256:DrT8qg1cP+6068axiACgib+3rrCMQAsav6PkAiVI0eQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'servera' (ED25519) to the list of known hosts.
production1@servera's password:
Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Mon Sep 28 20:17:44 2026
[production1@servera ~]$
```

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```
PasswordAuthentication no
-- INSERT --
```

```
[root@serverb ~]# vim /etc/ssh/sshd_config
[root@serverb ~]# systemctl restart sshd.service
[root@serverb ~]#
```

```
[production1@servera ~]$ ssh production2@serverb
production2@serverb: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[production1@servera ~]$
```

```
# Authentication:

#LoginGraceTime 2m
PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile /etc/.rht_authorized_keys .ssh/authorized_keys
```

```
[production1@servera ~]$ ssh production1@serverb
Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Fri Sep 25 21:18:58 2026 from 172.25.250.10
[production1@serverb ~]$
```

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```
[root@serverb ~]# vim /etc/ssh/sshd_config
[root@serverb ~]# exit
logout
[production1@serverb ~]$ exit
logout
Connection to serverb closed.
[production1@servera ~]$ exit
logout
[student@servera ~]$ exit
logout
Connection to servera closed.
student@workstation:~$
```

```
student@workstation:~$ lab grade ssh-review
🤖 Running: lab grade ssh-review
PASS    Verifying correct SSH key-based authentication configuration
        - The SSH key-based authentication configuration is correct.
PASS    Verifying prevention of root user SSH login to serverb
        - SSH login of root user is disabled
PASS    Verifying prevention of SSH password-based login to serverb
        - SSH password-based login on serverb is disabled
student@workstation:~$ lab finish ssh-review
🤖 Running: lab finish ssh-review
SUCCESS Restoring password authentication on serverb
SUCCESS Restoring original /etc/ssh/sshd_config on serverb
SUCCESS Deleting the production1 user on servera
SUCCESS Deleting the production1 user on serverb
SUCCESS Deleting the production2 user on serverb
SUCCESS Removing backup file
SUCCESS Disabling sshpass on servera
student@workstation:~$
```